

1. Modules Description

The proposed CA Examination Preparation System is organized into three primary modules. For the partial demo, the Student Module (with Login Page and Study Materials) has been fully implemented. The Faculty Module and Admin Module are partially developed and presented in view-only form.

1.1 Student Module (Implemented)

1.1.1 Login Page

The Login Page serves as the entry point for all users of the system. It authenticates the user credentials (email and password) against the database and directs the user to their respective dashboard based on role. Input validation is performed at both the frontend (Flutter) and backend (FastAPI) levels to prevent unauthorized access.

Key features of the Login Page include role-based redirection (Student, Faculty, Admin), error messaging for invalid credentials, and secure credential handling.

1.1.2 Study Materials

After successful login, the student can access subject-wise and chapter-wise study materials uploaded by the faculty. Materials are displayed in a structured list, organized by subject and chapter. Students can view, download, or read materials directly within the application.

The Study Material module is connected to the backend API which fetches records from the MySQL database and serves the appropriate content to the authenticated student.

1.2 Faculty Module (Partial — View Only)

The Faculty Module allows educators to manage content and monitor student activity. In the current partial implementation, the Faculty Dashboard view is presented, which includes options to upload study materials and create mock test questions. Full functionality is under development and will be completed in the final submission.

1.3 Admin Module (Partial — View Only)

The Admin Module provides system-wide management capabilities including user management and syllabus configuration. The Admin Dashboard view is presented at this stage. Full CRUD operations on users and subjects will be completed in the final phase.

2. Table Design

The system uses a relational MySQL database. The following tables are designed to support all system modules:

2.1 Users Table

Column Name	Data Type	Constraints	Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user identifier
name	VARCHAR(100)	NOT NULL	Full name of the user
email	VARCHAR(150)	UNIQUE, NOT NULL	Email used for login
password	VARCHAR(255)	NOT NULL	Hashed password
role	ENUM	NOT NULL	Student / Faculty / Admin
created_at	DATETIME	DEFAULT NOW()	Account creation timestamp

2.2 Subjects Table

Column Name	Data Type	Constraints	Description
subject_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique subject identifier
subject_name	VARCHAR(200)	NOT NULL	Name of the CA subject

2.3 Chapters Table

Column Name	Data Type	Constraints	Description
chapter_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique chapter identifier
subject_id	INT	FOREIGN KEY (Subjects)	Reference to parent subject
chapter_name	VARCHAR(200)	NOT NULL	Name of the chapter

2.4 Study Materials Table

Column Name	Data Type	Constraints	Description
material_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique material identifier
chapter_id	INT	FOREIGN KEY (Chapters)	Reference to chapter
title	VARCHAR(255)	NOT NULL	Title of the material
file_path	VARCHAR(500)	NOT NULL	Path to uploaded file
uploaded_by	INT	FOREIGN KEY (Users)	Faculty who uploaded

2.5 Mock Tests Table

Column Name	Data Type	Constraints	Description
test_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique test identifier

chapter_id	INT	FOREIGN KEY (Chapters)	Reference to chapter
total_marks	INT	NOT NULL	Total marks for the test
created_by	INT	FOREIGN KEY (Users)	Faculty who created

2.6 Results Table

Column Name	Data Type	Constraints	Description
result_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique result identifier
user_id	INT	FOREIGN KEY (Users)	Student who took the test
test_id	INT	FOREIGN KEY (MockTests)	Test reference
score	INT	NOT NULL	Marks obtained
test_date	DATETIME	DEFAULT NOW()	Date and time of attempt

3. Data Dictionary

The data dictionary provides a detailed description of all data elements used in the system, their types, sizes, and meanings.

Attribute	Table	Type	Size	Description
user_id	Users	INT	11	Auto-generated unique user ID
name	Users	VARCHAR	100	Full name of the registered user
email	Users	VARCHAR	150	Unique email for login authentication
password	Users	VARCHAR	255	Bcrypt-hashed password string
role	Users	ENUM	-	Defines access: Student/Faculty/Admin
subject_id	Subjects	INT	11	Auto-generated unique subject ID
subject_name	Subjects	VARCHAR	200	Name of the CA Foundation subject
chapter_id	Chapters	INT	11	Auto-generated unique chapter ID
chapter_name	Chapters	VARCHAR	200	Name of the chapter within a subject
material_id	StudyMaterial	INT	11	Unique ID for each study material
title	StudyMaterial	VARCHAR	255	Title/name of the study material
file_path	StudyMaterial	VARCHAR	500	Server path to stored file
test_id	MockTests	INT	11	Unique ID for each mock test
total_marks	MockTests	INT	4	Total marks allocated to the test
result_id	Results	INT	11	Unique ID for each test result
score	Results	INT	4	Marks obtained by the student
test_date	Results	DATETIME	-	Timestamp of test attempt

4. Program Design Language (PDL)

The following pseudo-code describes the core logic of the implemented modules: Login and Study Material access.

4.1 PDL — User Login

```
BEGIN LOGIN
  INPUT: email, password
  IF email IS EMPTY OR password IS EMPTY THEN
    DISPLAY error: 'Fields cannot be empty'
    RETURN
  END IF
  SEND POST request to /api/login with {email, password}
  IF response.status == 200 THEN
    STORE token and role in local session
    IF role == 'student' THEN
      NAVIGATE to Student Dashboard
    ELSE IF role == 'faculty' THEN
      NAVIGATE to Faculty Dashboard
    ELSE IF role == 'admin' THEN
      NAVIGATE to Admin Dashboard
    END IF
  ELSE
    DISPLAY error: 'Invalid credentials'
  END IF
END LOGIN
```

4.2 PDL — Fetch Study Materials

```
BEGIN FETCH_STUDY_MATERIALS
  INPUT: subject_id (selected by student)
  SEND GET request to /api/materials?subject_id={subject_id}
  WITH Authorization header = student_token
  IF response.status == 200 THEN
    materials_list = response.data
    FOR EACH material IN materials_list DO
      DISPLAY material.title
      DISPLAY material.chapter_name
      SHOW download/view button
    END FOR
  ELSE
    DISPLAY error: 'Unable to load materials'
  END IF
END FETCH_STUDY_MATERIALS
```

4.3 PDL — Backend: Authenticate User (FastAPI)

```
BEGIN API_LOGIN_HANDLER
  INPUT: email, password (from request body)
  QUERY Users TABLE WHERE email = input.email
  IF record NOT FOUND THEN
    RETURN HTTP 401: 'User not found'
  END IF
  IF bcrypt.verify(input.password, record.password) == FALSE THEN
    RETURN HTTP 401: 'Invalid password'
  END IF
  token = JWT.generate({ user_id, role }, SECRET_KEY)
  RETURN HTTP 200: { token, role, name }
END API_LOGIN_HANDLER
```

5. Feasibility Study

A feasibility study was conducted to determine whether the proposed CA Examination Preparation System is technically, operationally, and economically viable.

5.1 Technical Feasibility

The system is technically feasible as all chosen technologies are well-established, open-source, and widely supported. Flutter enables cross-platform development for both Android and iOS from a single codebase, reducing development effort. FastAPI (Python) is a lightweight and high-performance framework for building RESTful APIs with minimal overhead. MySQL is a mature and reliable relational database management system suitable for storing structured data such as user records, question banks, and performance results. All development tools (VS Code, Postman, GitHub) are freely available.

Technology	Role	Feasibility
React	Frontend / Mobile UI	High — cross-platform, free, large community
FastAPI (Python)	Backend API	High — fast, lightweight, well-documented
MySQL	Database	High — mature RDBMS, free, widely used
VS Code	IDE	High — free, extensible
GitHub	Version Control	High — free, industry standard

5.2 Operational Feasibility

The system is operationally feasible. CA Foundation students routinely use smartphones and are comfortable with mobile applications. The proposed system consolidates study materials and mock tests in one place, simplifying the preparation process. Faculty members can manage content through an intuitive interface without requiring advanced technical knowledge. Admin users can manage the system through a dedicated dashboard. The system is designed to minimize learning curve and maximize adoption.

The role-based access control ensures that each user type (Student, Faculty, Admin) only sees features relevant to them, making the interface clean and easy to use.

5.3 Economic Feasibility

The system is economically feasible for an academic project. All technologies used are open-source and free of licensing fees. The development is carried out using personal hardware (laptop with minimum 4 GB RAM as specified). Hosting the application during development is done locally, with deployment to a shared or cloud server planned for the final release. The estimated cost is limited to internet charges and domain/hosting fees, both of which are minimal.

6. Journal Paper

A Cross-Platform CA Examination Preparation System

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Abstract

The Chartered Accountancy (CA) examination is one of the most challenging professional examinations in India, requiring students to study multiple subjects and continuously assess their performance. Existing preparation tools are fragmented across different platforms, resulting in unorganized and inefficient study practices. This paper presents the design and partial implementation of a cross-platform mobile application for CA Foundation exam preparation, developed using Flutter, FastAPI, and MySQL. The system integrates study materials, mock tests, and performance analysis in a unified platform, accessible to students, faculty, and administrators through role-based access control. The partial demo demonstrates the working Login module and Study Material access for students. Results indicate that the proposed system effectively consolidates preparation resources and improves the study workflow for CA Foundation candidates.

Keywords

CA Examination, Flutter, FastAPI, MySQL, E-Learning, Mock Tests, Study Materials, Role-Based Access, Cross-Platform Application

1. Introduction

The Chartered Accountancy (CA) Foundation examination conducted by the Institute of Chartered Accountants of India (ICAI) covers four papers: Principles and Practice of Accounting, Business Laws and Business Correspondence and Reporting, Business Mathematics, Logical Reasoning and Statistics, and Business Economics and Business and Commercial Knowledge. Students preparing for this examination often rely on multiple disjointed platforms — video lectures on one service, mock tests on another, and notes in printed form — leading to an unorganized study experience.

Mobile and web-based learning platforms have become a significant aid in examination preparation [1][2]. Research in e-learning systems highlights the importance of integrating study content, assessment, and performance feedback within a single interface [3]. However, no dedicated, lightweight application exists specifically for CA Foundation students that combines all these features.

This paper describes the architecture and development of a cross-platform CA examination preparation system. The system uses Flutter for the frontend, FastAPI for the backend, and MySQL for data management. The current implementation covers the Login module and Study Material module for students, with partial views for Faculty and Admin modules.

2. Related Work

Several studies have examined mobile-based learning systems. Ally [1] highlighted that mobile learning offers flexibility and accessibility for students. Kukulska-Hulme et al. [2] examined mobile applications for examination preparation and found that integrated systems perform better than single-purpose tools. Existing CA preparation platforms such as ICAI Study Material Portal and third-party mock test apps address some aspects of preparation but lack the integration of content, testing, and analytics in one application [3].

Research by Alavi and Leidner [4] on knowledge management systems in education emphasizes the role of structured content delivery and timely feedback. The proposed system addresses these needs by providing organized subject-wise materials and chapter-wise test access within a single application.

3. System Architecture

The system follows a three-tier client-server architecture. The Flutter frontend communicates with the FastAPI backend through RESTful API calls. The backend processes business logic and interacts with the MySQL database. Role-based access control ensures that students, faculty, and administrators receive appropriate interface views.

The three main modules are: (i) Student Module — allows students to login, view study materials, attend mock tests, and view results; (ii) Faculty Module — allows faculty to upload study materials and create mock tests; and (iii) Admin Module — manages users, subjects, and system-level configurations. In the partial demo, the Student Module (Login and Study Materials) is fully implemented, while Faculty and Admin modules are presented as view-only dashboards.

4. Implementation

The Login page was implemented using Flutter's `StatefulWidget` with form validation. The backend login endpoint in FastAPI validates user credentials against bcrypt-hashed passwords stored in MySQL. On successful authentication, a JWT token is issued and stored in the client session for subsequent API calls.

The Study Material module fetches subject-wise and chapter-wise materials from the backend via a secured GET endpoint. The retrieved data is displayed in a `ListView` widget in Flutter, allowing students to browse and access uploaded documents.

5. Results and Discussion

The partial demo demonstrates that the login and study material access modules function as expected. User authentication is secure, with role-based navigation correctly directing users to their respective dashboards. Study materials are retrieved dynamically from the database and displayed in an organized manner.

The Faculty and Admin module dashboards are rendered correctly in the UI, providing a foundation for full implementation in the next phase. The incremental development

methodology adopted allowed early testing and validation of core modules before extending to more complex features.

6. Conclusion

This paper presented the partial implementation of a cross-platform CA examination preparation system. The Login and Study Material modules have been successfully developed and demonstrated. The system addresses the key gap identified in existing platforms by integrating study content and test access in a single mobile application. Future work includes completing the Mock Test module, Performance Analysis module, and full Faculty and Admin management features.

References

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